

Equivalence of Functional Dependency Sets

Checking Coverage • Attribute Closure Method • Solved Exam Examples

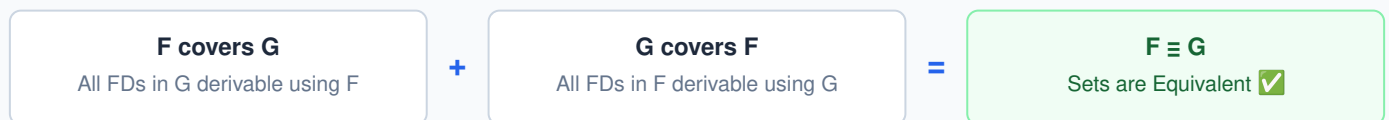
In Database Management Systems, two different-looking sets of functional dependencies (FDs) may actually represent the exact same logical constraints on a relation. Determining whether two sets are equivalent is a fundamental topic for university exams, GATE, and database normalization.

★ Definition of Equivalence

Two functional dependency sets F and G are **equivalent** (written as $F \equiv G$) if and only if **F covers G** AND **G covers F**. Mathematically, $F \equiv G \iff F^+ = G^+$.

1. What Does "F Covers G" Mean?

To say that **F covers G** ($F \Rightarrow G$) means that *every functional dependency present in G can be derived from F*.



2. Exam Method Using Attribute Closure ★

To check equivalence between set F and set G in exams, perform two coverage checks:

Step 1: Check if F covers G ($F \Rightarrow G$)

1. For each dependency $X \rightarrow Y$ in set G , take the LHS attribute(s) X .
2. Calculate the attribute closure X^+ using the FDs in set F .
3. Verify whether $Y \in X^+$.
4. If **every** FD in G passes, then **F covers G**.

Step 2: Check if G covers F ($G \Rightarrow F$)

1. For each dependency $X \rightarrow Y$ in set F , take the LHS attribute(s) X .
2. Calculate the attribute closure X^+ using the FDs in set G .
3. Verify whether $Y \in X^+$.
4. If **every** FD in F passes, then **G covers F**.

⚠ Common Student Trap: Which Set's Closure to Use?

- When testing **F covers G**: Take FDs from **G**, but compute closure using **F**.
- When testing **G covers F**: Take FDs from **F**, but compute closure using **G**.

Memory Trick: "What you are covering $\rightarrow$ take its FD. Who is covering $\rightarrow$ use its closure."

3. Solved Examples

Example 1: Proving Equivalent Sets ($F \equiv G$)

Given the two dependency sets:

F = { $A \rightarrow B$, $B \rightarrow C$, $A \rightarrow C$ }

G = { $A \rightarrow B$, $B \rightarrow C$ }

Step 1: Check if F covers G ($F \Rightarrow G$)

We test each FD in **G** using closure under **F**:

- Test $A \rightarrow B$: Compute A^+ using $F \rightarrow A^+ = \{A, B, C\}$. Since $B \in A^+$ **Pass**
- Test $B \rightarrow C$: Compute B^+ using $F \rightarrow B^+ = \{B, C\}$. Since $C \in B^+$ **Pass**

Result: **F covers G**

Step 2: Check if G covers F ($G \Rightarrow F$)

We test each FD in **F** using closure under **G**:

- Test $A \rightarrow B$: Directly present in **G** **Pass**
- Test $B \rightarrow C$: Directly present in **G** **Pass**
- Test $A \rightarrow C$: Compute A^+ using **G** $\rightarrow$ Start: $\{A\} \rightarrow$ via $A \rightarrow B$: $\{A, B\} \rightarrow$ via $B \rightarrow C$: $\{A, B, C\}$. Since $C \in A^+$ **Pass**

Result: **G covers F**

Conclusion: Since both directions hold true, $F \equiv G$

Example 2: Proving Non-Equivalent Sets ($F \neq G$)

Given the two dependency sets:

F = { $AB \rightarrow CD$, $B \rightarrow C$, $C \rightarrow D$ }

G = { $AB \rightarrow C$, $AB \rightarrow D$, $C \rightarrow D$ }

Step 1: Check if F covers G ($F \Rightarrow G$)

- Test $AB \rightarrow C$: $(AB)^+$ using $F = \{A, B, C, D\} \rightarrow C \in (AB)^+$ **Pass**

- Test $AB \rightarrow D$: $(AB)^+$ using $F = \{A, B, C, D\} \rightarrow D \in (AB)^+$ **Pass**
- Test $C \rightarrow D$: Exists directly in F **Pass**

Result: **F covers G**

Step 2: Check if G covers F ($G \Rightarrow F$)

- Test $AB \rightarrow CD$: $(AB)^+$ using $G = \{A, B, C, D\} \rightarrow$ **Pass**
- Test $B \rightarrow C$: Compute B^+ using G :
 - Start: $B^+ = \{B\}$
 - Can we apply $AB \rightarrow C$? No, attribute A is missing.
 - Can we apply $C \rightarrow D$? No, attribute C is missing.
 - Final $B^+ = \{B\}$.
- Since $C \notin B^+ \rightarrow$ **Failed! $B \rightarrow C$ cannot be derived using G.**

Result: **G does NOT cover F**

Conclusion: Since G does not cover F, $F \neq G$

4. Equivalence vs. Minimal Cover

Do not confuse **Equivalence** with **Minimal Cover**. They address different aspects of functional dependencies:

Feature	Equivalence Check	Minimal Cover
Input	Two separate FD sets (F and G)	A single FD set (F)
Goal	Determines if both sets express the same constraints	Produces an irreducible, minimal representation
Operations Used	2-way coverage checking using attribute closure	RHS decomposition, redundancy removal, LHS minimization
Key Property	$F \equiv G$	$F_{min} \equiv F$ (A minimal cover is always equivalent to original set)

Equivalence Decision Rule:

(F covers G = YES) & (G covers F = YES) $\Rightarrow F \equiv G$

One-Line Exam Memory Trick

“To prove equivalence, prove coverage in BOTH directions using the covering set's closure.”